

Prompting Fundamentals on Real Tasks

Task

Build a CRUD API using FastAPI

This task was selected from my Backend AI internship assignment. The objective was to improve a simple prompt through multiple iterations using different prompt engineering techniques and observe how each refinement affected the quality of the AI-generated response.

Baseline Prompt

Prompt

Build a CRUD API using FastAPI.

Representative Output

The response immediately generated a complete CRUD API with FastAPI using an in-memory database. It included the project structure, installation commands, API endpoints, and complete source code. While the generated solution was functional, it focused primarily on providing the implementation instead of explaining the backend concepts.

Notes

- **Technique Used**
None (Naive Prompt)
 - **What changed in the prompt?**
This was my original one-line prompt without any additional instructions or context.
 - **What improved in the output?**
The AI generated a complete working CRUD API that could be executed immediately.
 - **What still failed?**
The response assumed I already understood backend development and focused more on code than learning.
 - **What would I try next?**
Assign the AI the role of a mentor so it teaches instead of simply generating code.
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Version 1

Technique Used

Role Assignment

Prompt

You are an experienced backend developer and mentor.

Help me build a CRUD API using FastAPI. Explain everything in simple language suitable for a beginner.

Representative Output

The response became more educational. Instead of directly providing the code, it first explained CRUD operations, why FastAPI is used, Pydantic models, and the purpose of each API endpoint before presenting the implementation.

Notes

- **What changed in the prompt?**
I assigned the AI the role of an experienced backend developer and mentor.
 - **What improved in the output?**
The explanation became beginner-friendly and focused on teaching the concepts instead of only generating code.
 - **What still failed?**
The explanation still assumed its own teaching style and did not consider my learning background.
 - **What would I try next?**
Provide my background and learning objective.
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Version 2

Technique Used

Context & Motivation

Prompt

You are an experienced backend developer and mentor.

I am a Computer Science student completing the FlyRank Backend AI internship. I am learning backend development for the first time and need to build a CRUD API using FastAPI as part of my internship assignment.

Please explain every concept in simple language, focus on why each step is needed, and help me understand how the backend processes client requests instead of only giving me the final code.

Representative Output

The response explained the backend request lifecycle, how client requests are processed, why FastAPI validates data before executing a function, and why separate Pydantic models are used for creating, updating, and returning data.

Notes

- **What changed in the prompt?**
I added my learning background, internship context, and explained why I needed to understand the backend instead of only generating code.
- **What improved in the output?**

The AI became much more educational. It explained why backend concepts exist and how requests travel through the application before returning a response.

- **What still failed?**

Although the explanations became much more detailed, the response still followed a generic teaching structure.

- **What would I try next?**

Provide an example of the explanation format I expect.

Version 3

Technique Used

Few-shot Example

Prompt

You are an experienced backend developer and mentor.

I am a Computer Science student completing the FlyRank Backend AI internship. I am learning backend development for the first time and need to build a CRUD API using FastAPI.

I prefer explanations like this:

Concept:

A REST API is a communication bridge between the client and the server.

Simple Explanation:

When a client sends a request, the backend processes it, performs the required operation, and returns the appropriate response.

Please explain every concept in this same format:

- Concept
- Simple Explanation
- Why it Matters
- Small Real-world Example

Then build the CRUD API step by step.

Representative Output

The AI consistently explained every concept using the requested format: Concept → Simple Explanation → Why it Matters → Real-world Example, making the lessons easier to follow.

Notes

- **What changed in the prompt?**

I provided an example of the explanation style before asking the AI to teach.

- **What improved in the output?**

The explanations became much more consistent and easier to understand because every concept followed the same structure.

- **What still failed?**

Although the explanation style improved, the response was still quite lengthy and not divided into implementation phases.

- **What would I try next?**

Specify the desired output structure.

Version 4

Technique Used

Output Structure

Prompt

You are an experienced backend developer and mentor.

I am a Computer Science student completing the FlyRank Backend AI internship. I am learning backend development for the first time and need to build a CRUD API using FastAPI.

I prefer explanations in this format:

Concept

Simple Explanation

Why it Matters

Real-world Example

Now organize your entire response using this structure:

1. Project Overview
2. Prerequisites
3. Concepts
4. Project Structure
5. Step-by-Step Implementation
6. Complete Code
7. API Testing using Swagger
8. Common Beginner Mistakes
9. Next Learning Steps

Representative Output

The response became well-organized with clearly separated sections such as Project Overview, Concepts, Implementation, Swagger Testing, Common Mistakes, and Next Learning Steps.

Notes

- **What changed in the prompt?**

I specified the exact structure I wanted the AI to follow.

- **What improved in the output?**

The information became much easier to navigate and study because every topic was presented under its own heading.

- **What still failed?**

The implementation was still delivered in large sections, which could overwhelm a beginner.

- **What would I try next?**

Break the project into smaller learning steps.

Version 5

Technique Used

Step Decomposition

Prompt

You are an experienced Backend Developer, FastAPI expert, and technical mentor.

I am a Computer Science student currently completing the FlyRank Backend AI Internship. I am learning backend development for the first time.

My Goal:

I do not want to simply finish the assignment. I want to deeply understand backend development so that I can confidently build CRUD APIs on my own.

Teach me by introducing only one concept at a time, giving one small coding task, reviewing my work, and waiting for my confirmation before continuing.

Representative Output

The AI introduced only one lesson, explained the FastAPI application object, assigned a small coding task, described the expected outcome, highlighted common beginner mistakes, and waited for my response before continuing.

Notes

- **What changed in the prompt?**

I instructed the AI to teach one concept at a time using an interactive mentoring workflow.

- **What improved in the output?**

The response became much more interactive. Instead of providing the complete project, it guided me through one lesson at a time, making the learning process easier and more practical.

- **What still failed?**

Completing the full project requires multiple interactions, making the learning process longer. However, this approach provides a deeper understanding of the concepts.

- **What would I try next?**

Continue using the same mentoring approach for more advanced backend topics such as authentication, databases, and testing.

Claude vs ChatGPT Comparison

Aspect	Claude	ChatGPT
Tone	Very patient and conversational, like an internship mentor.	Friendly, structured, and concise while maintaining a mentoring style.
Accuracy	Explained backend concepts correctly with detailed conceptual guidance.	Also explained concepts accurately and provided practical implementation guidance.
Structure	Followed the requested lesson structure exactly.	Closely followed the requested structure with clear formatting and logical flow.
Failure Points	Responses were more detailed, making them longer to read.	Responses were slightly shorter and more concise but introduced fewer conceptual details in the first lesson.
Overall	Better for learners who want detailed conceptual understanding.	Better for learners who prefer concise explanations with a structured teaching approach.

Final Reusable Prompt Template

You are an experienced developer and mentor.

I am a beginner learning a new technology.

My goal is to understand the concepts deeply instead of only completing the project.

Assume I know only the basics of programming.

Teach me by following this workflow:

1. Explain one concept at a time.
2. Give one small practical task.
3. Wait for my solution.
4. Review my work.
5. Explain my mistakes.
6. Move to the next concept only after I understand the current one.

For every lesson, use this structure:

- Objective
- Concept
- Simple Explanation
- Why it Matters
- Real-world Example

- Behind the Scenes
- Small Coding Task
- Expected Outcome
- Common Mistakes

Use simple language, explain why before how, avoid unnecessary jargon, and never assume prior knowledge. Do not provide the complete project immediately. Guide me step by step until the project is complete.

Reflection

This assignment helped me understand that effective prompt engineering is an iterative process rather than writing one perfect prompt. By improving a simple prompt step by step using techniques such as role assignment, context, few-shot examples, output structure, and step decomposition, I observed how each refinement changed the quality of the AI-generated response. I also learned that different AI models have different strengths. Claude provided more detailed conceptual explanations, while ChatGPT delivered concise and well-structured responses. Overall, this assignment improved my ability to design prompts that produce clearer, more useful, and more educational outputs.